When the Average Lies: Validating Signals Before Trusting a Metric



A statistical analysis of Google Merch Shop users' spending behavior in August 2017

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Do repeat visitors generate meaningfully higher revenue than single-visit buyers?

Assumption

In SEO and growth marketing, users who visit multiple times are assumed to have high buying intent.

So what?

If true: justifies prioritizing retention and remarketing budget.

If false: resources are better allocated elsewhere

Data and methodology

1 **BigQuery public dataset**

2,256 rows and three columns (visitorId, visitNumber, trxRev) were queried from the GA-UA Public Dataset (August 2017)

2 **Data aggregation**

Aggregated to four columns (2293 rows). Grouped by fullVisitorId inside BigQuery before exported to Python. Columns name: jumlah_visit, visitor_type

3 **Descriptive statistics**

Initial analysis by descriptive stat (Sum, Mean, Count)

4 **EDA & hypothesis test**

Executed EDA on data dispersion and ran a Mann-Whitney U test

A single outlier inflated the average revenue of "repeat visitors" by 62%. Statistical validation revealed the initial signal wasn't reliable.

13x gap

Initial analysis reflected on average **repeat visitors spent 13x more** than single-visit buyers ($22 vs $1.71), **seemingly agree to the original hypothesis**.

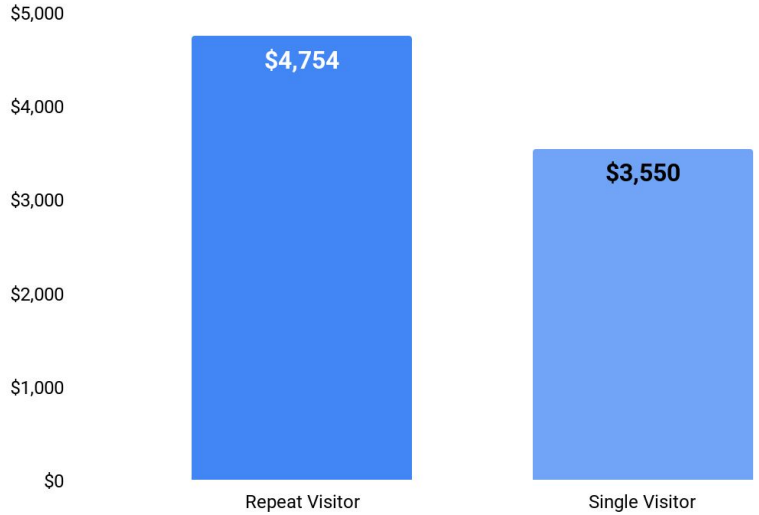
The shift

Whale users inflated the averages. **Median uncovers different pattern** where Single visitors tend to spent higher ($48 vs. $34)

Stat test

A Mann-Whitney U test (p = 0.533) found **no statistically significant difference between groups.** The observed gap could plausibly be due to chance

Initial analysis suggests that repeat visitors were 34% more lucrative. Their Average Order Value (AOV) was around $22.



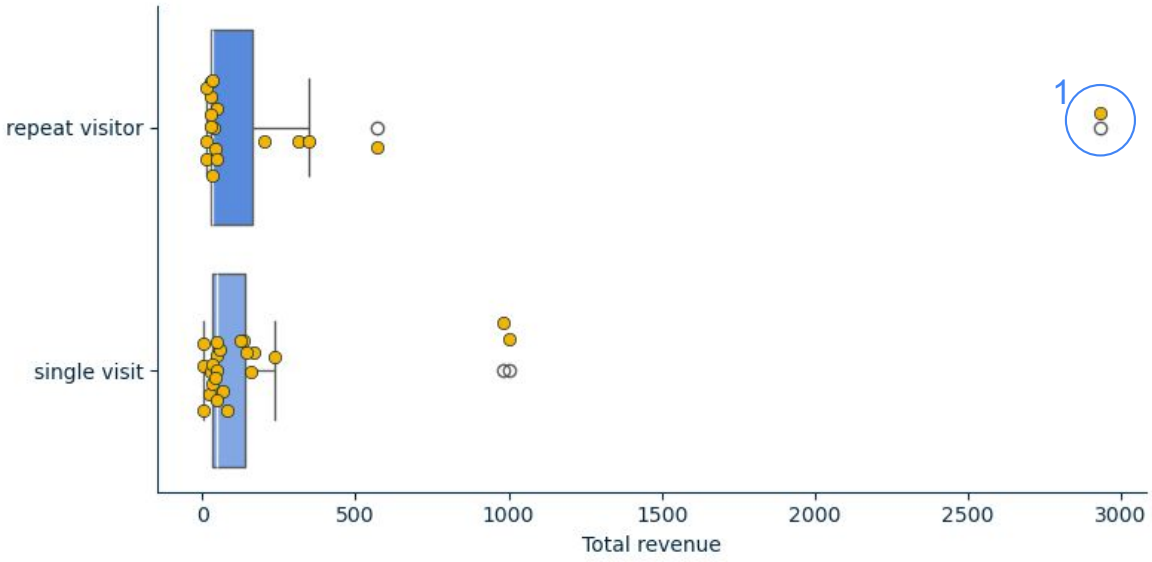
The illusion: **from 42 transactions** in August (18 repeat visitors transaction and 24 single visitor transactions), **both sum and mean showed single visitor group spend more than its counterpart.**

The risk: If the analysis stops here, the business might aggressively shifts marketing campaign budget to retargeting.

So what? The data distribution must be checked to reveal its true pattern before committing budget allocation.

The median suggests single visitors possess higher spending power. Opposite to the initial analysis.

Both groups reflect a **positively skewed distribution,** majority of visitors shopped in small value. 1 out of 18 Repeat Visitor contributed to 62% of its total revenue[1]. **Distorting its mean and sum.**



Distorted mean data

Repeat visitor group wins
$22 vs. $1.7

Adjusted median data

Single visitor group wins
$48 vs. $34

But is this a valid finding? Or pure chance?

A Mann-Whitney U Test confirms the median spending difference is insufficient to prove.

The method

A two-sided Mann-Whitney U Test is used due to the non-parametric data and low population.

The hypotheses

H_0: Repeat visitor = Single visit ⟶ the difference is happened due to mere luck

H_1: Repeat visitor ≠ Single visit ⟶ One group is superior than the other, not due to luck.

```
from scipy.stats import mannwhitneyu
group_single = df_trx[df_trx['visitor_type'] == 'single visit']['total_revenue']
group_repeat = df_trx[df_trx['visitor_type'] == 'repeat visitor']['total_revenue']

statistic, p_value = mannwhitneyu(group_single, group_repeat, alternative='two-sided')
print(statistic, p_value)
```

P-value = 0.533 (>0.05 α)
U statistic = 241

Conclusion

The test **fails to prove** a spending gap between the two groups.

The difference mentioned earlier **happened because of chance,** and does not represent a true pattern of Google Merch Shop's Visitors.

Key takeaway: always evaluate data distribution before executing budget-impacting business decisions.

Evaluate distribution

A single transaction skewed the trend, where 62% of repeat visitors' revenue came from one buyer. While mean can be distorted, **checking data distribution is a way to stay truthful.**

Limitation

Limitation is present in this research, such as **small sample size** (date range), which results in low statistical power.

What's next?

Extend the sample date range, and re-run the hypothesis test.

Appendix

Github link
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